

Notepad Application Documentation

Software Configuration Management - Lab

(SE 503)

Course : Nabeela Ashraf
Instructor

Submitted By : Muhammad Ahmad
Saqib Ali Tahir
Muhammad Ahmad
Usman Abid

Ag-no : 2022-ag-8042
2022-ag-8088
2022-ag-8049
2022-ag-8085



University of Agriculture, Fsd

Overview

This project is a text editor application developed using Python and the Tkinter GUI library. The application provides basic text editing functionalities such as creating, opening, saving, and editing text files. Additional features include font customization, undo/redo actions, and support for keyboard shortcuts.

Technologies Used

1. Python

- **Description:** The programming language used to implement the application.
- **Reason for Use:** Python is versatile and includes robust libraries for GUI development, making it an excellent choice for desktop applications.

2. Tkinter

- **Description:** A built-in Python library for creating graphical user interfaces (GUIs).
- **Components Used:**
 - **Text:** A widget for editing and displaying text.
 - **Menu:** Provides a menu bar for various operations.
 - **Scrollbar:** Adds scroll functionality to the text widget.
 - **Messagebox:** Displays dialog boxes for user interactions.
 - **FileDialog:** Allows file selection for opening and saving.
 - **Colorchooser:** Provides a color selection dialog.
 - **Toplevel** and **Combobox** (from ttk): Used for advanced font selection dialogs.

3. ttk Module

- **Description:** Part of Tkinter for modern-looking widgets.
- **Usage:** Used to enhance the UI with widgets like **Combobox** for font family selection.

Application Structure

The application is encapsulated in the `Notepad` class, which handles initialization, GUI setup, and event handling.

Class: Notepad

Initialization (`__init__`)

- Sets up the main application window with title, size, and default settings.
- Initializes variables:
 - `file`: Tracks the current file path.
 - `text_changed`: Tracks if the text area content has been modified.
 - `current_font_family`, `current_font_size`, and `text_color`: Manage text styling.
- Creates and configures the text editing area, scrollbars, and menus.

GUI Components

1. **Text Widget**
 - Displays the main text area for editing.
 - Configured with word wrapping, undo capability, and default font settings.
2. **Scrollbar**
 - Attached to the text widget for vertical scrolling.
3. **Menu Bar**
 - Contains the following menus:
 - **File Menu**: Options to create, open, save, save as, and exit.
 - **Edit Menu**: Text editing commands such as undo, cut, copy, paste, and delete.
 - **Format Menu**: Options to change font family, size, and color.
 - **Help Menu**: Displays an About dialog.

Functions and Methods

File Operations

- `new_file()`: Clears the text area and resets the application state.
- `open_file()`: Opens a file dialog, reads the selected file, and displays its contents.
- `save_file()`: Saves the current text to the existing file.
- `save_as_file()`: Opens a save dialog and writes the text to a new file.

- **exit_notepad()**: Prompts to save changes if necessary and closes the application.
- **prompt_save_changes()**: Handles the save confirmation dialog when changes are unsaved.

Text Editing

- **undo()**: Reverts the last text change.
- **cut()**: Cuts the selected text to the clipboard.
- **copy()**: Copies the selected text to the clipboard.
- **paste()**: Pastes the clipboard content at the cursor.
- **delete_text()**: Deletes the selected text.

Formatting

- **change_font_family()**: Opens a dialog to select a new font family and applies it.
- **change_font_size()**: Prompts for a new font size and updates the text area.
- **change_font_color()**: Opens a color picker and changes the text color.

Utility

- **on_text_change()**: Tracks modifications in the text area and updates the window title to indicate unsaved changes.
- **show_about()**: Displays information about the application.

Event Binding

- **<<Modified>>**: Tracks changes in the text area.

How It Works

1. The application starts by initializing the **Notepad** class and creating the main tkinter window.
2. Users can perform file operations using the File menu.
3. Text editing is supported through the Edit menu and keyboard shortcuts.
4. Text formatting options allow customization of font family, size, and color.
5. The application ensures unsaved changes are prompted before closing or opening a new file.

Key Features

- **File Management:** Create, open, save, and save as text files.
- **Text Editing:** Basic text manipulation commands.
- **Formatting Options:** Change font family, size, and color.
- **Undo/Redo:** Support for undoing changes.
- **Customizable Appearance:** Font and color adjustments.
- **User-Friendly Alerts:** Notifications for unsaved changes and errors.

Usage

1. Run the script to start the application.
2. Use the menu bar for operations such as opening a file, editing text, and formatting.
3. Keyboard shortcuts, like `Ctrl+S`, `Ctrl+C` etc enhance usability for frequent actions.

Dependencies

- Python 3.x
- `tkinter` library (standard with Python installation)

Notes

This Notepad application serves as a foundation for text editors and can be extended with additional features like search/replace, rich text formatting, or spell-checking.

References

- <https://www.geeksforgeeks.org/make-notepad-using-tkinter>
- <https://youtu.be/uf1IAjxmsD4?si=ILRrm1X8tmBFm5cX>
- chatgpt.com